With its capital at the end, GPT-2 finds the country anywhere and rests less

GPT-2 small, 200 random rows × 11 country–capital pairs per filler type. The final token T is the country's capital, the same in both versions of each pair. The versions differ only at position τ: the country, or the row's own token. Δ = attention from T to τ, country − original, at every distance $T - \tau = 1 \ldots 31$. Control: the same with a wrong capital at T (France … Tokyo).

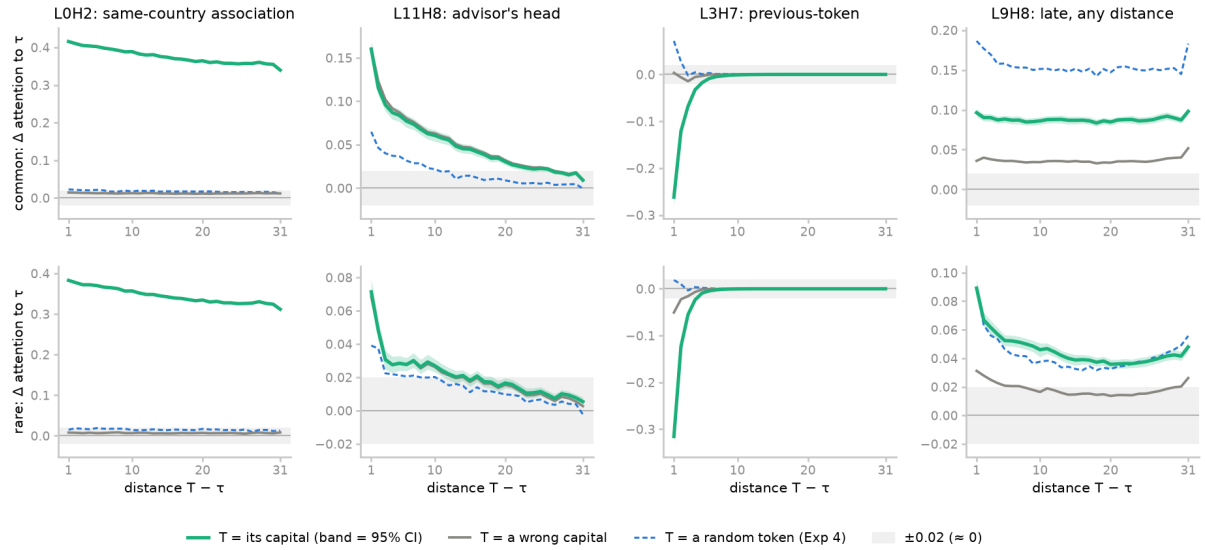
original BOS she sc … 26 random tokens … age dis Paris **country** BOS she sc … 26 random tokens … France dis Paris (τ = 30, distance 2)

Conclusion (common / rare filler)

- **The matching capital lowers late-layer resting at every distance.** Late (layers 6–11) attention on BOS drops by **0.031** / **0.017** at distance 1 and by 0.033 / 0.028 at distances 16–31, against a wrong capital. It needs the country (a placebo word gives ≈ 0), and the freed attention does not go to the country. It holds for 10 of the 11 pairs; Canada is the exception.
- **Early heads link same-country tokens at any distance.** With its own capital at T, L0H2 puts **0.43** / **0.40** of its attention on the country (0.38 / 0.35 twenty tokens back), against 0.03 with a wrong capital. L4H10, L3H0 and L1H11 do the same. They are not specific to "capital of": demonyms, non-capital cities and the reverse direction work too. In layer 0 this is a fixed token-pair association.
- **The advisor's L11H8 is boosted by a place-name query, not by the fact.** Its $\Delta(1)$ is +0.160 / +0.071 with its capital, the same with a wrong capital, and +0.065 / +0.039 with a random token. A non-capital city boosts it as much.
- **Previous-token heads look away from an adjacent same-country token**, toward the capital itself and the token two back. For example, L3H7 drops from 0.51 to 0.24 on France, and even L4H11 drops from 0.98 to 0.91 / 0.86.

1 · Four heads, three final tokens

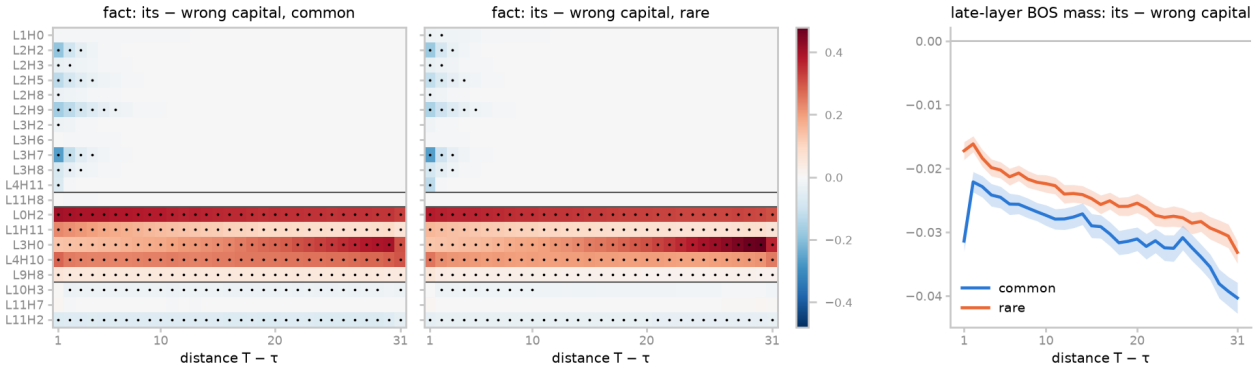
top: common filler; bottom: rare; grey band = $|\Delta| < 0.02$



L0H2: only the matching capital, flat with distance. L11H8: any capital. L3H7: drops only for a matching adjacent country. L9H8: a wrong capital suppresses it (a random final token gives as much or more).

2 · Where the matching capital matters

dots = BH $q \leq 0.05$ and $|\Delta| \geq 0.02$



Rows: previous-token heads; L11H8; same-country heads and L9H8; L10H3, L11H7, L11H2. L11H2 and, more weakly, L10H3 attend *less* to the country under its own capital.

Pre-registered (exp4/PREDICTIONS_capital.md): F1, F2 and F5 pass. F3, F4 and F6 fail, F6 narrowly (L1H0 at distance 9). All checks pass, and an independent recomputation matched. Probes: 30 rows. Random final token: exp4_onepager.pdf. Interactive: country slider, per-head viewer.